

1 electrolyte proportion near the SE suppresses interfacial side reactions and enhances
2 ion permeation. Such gradient structures maintain a three-dimensional continuous dual
3 channel for ions and electrons while effectively mitigating local polarization and
4 interfacial instability, demonstrating significant advantages under high loading and long
5 cycling conditions.
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10 Beyond composition ratios, the fabrication process of composite cathodes
11 profoundly influences the microstructure and electrochemical behavior at the cathode
12 interface (**Figure 10a**) ^[150]. Currently, composite cathodes predominantly rely on
13 mechanical mixing processes, including manual grinding or ball milling pretreatment.
14 Compared to simple manual mixing, controlled mechanical mixing significantly
15 improves the uniform distribution of sulfide SE on CAM particle surfaces, reduces
16 agglomeration, and realizes a quasi-coated structure to some extent, thereby
17 constructing a more continuous ion/electron transport network. Even under low
18 electrolyte loading conditions, such structures maintain low interfacial impedance and
19 excellent cycling stability. However, traditional mechanical mixing processes still have
20 inherent limitations. On one hand, conventional grinding struggles to achieve uniform
21 mixing at the nanoscale and results in a limited actual contact area between active
22 particles and electrolyte. On the other hand, high-energy ball milling can refine particle
23 size, yet it may cause damage to the cathode crystal structure, increase interface defects,
24 and exacerbate side reactions.
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41 Applying external pressure is another crucial method for enhancing solid-solid
42 interface contact (**Figure 10b**) ^[151]. As compaction pressure increases, the porosity of
43 the composite cathode decreases significantly while the actual contact area between the
44 CAM and sulfide SE continuously expands. This simultaneously reduces both the
45 interfacial charge transfer impedance and the bulk electrolyte resistance. However,
46 excessive pressure may increase electrolyte channel tortuosity and introduce
47 anisotropic contact structures within the composite cathodes. Specifically, contact
48 along the pressure direction improves significantly, while weakly connected regions
49 dominated by pores persist laterally. This anisotropic structure can still become a
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bottleneck limiting performance enhancement under high-rate or thick electrode conditions. Regarding synergistic optimization of composition and structure, the combination of gradient cathode structures with single-crystal CAMs demonstrates significant complementary effects (**Figure 10c**) ^[152]. Single-crystal CAMs with uniform size and low grain boundary density can substantially reduce stress concentration and crack initiation tendencies during cycling, thereby enhancing the long-term cycling stability of gradient structures (**Figure 10d**) ^[153]. Simultaneously, the short lithium-ion diffusion paths and uniform electrochemical reaction interfaces further amplify the advantages of gradient design in mitigating ion transport limitations and interface polarization. Integrating single-crystal CAM with gradient composition distribution enables the simultaneous achievement of high-capacity utilization and excellent cycling stability under high loading conditions.

The stability of the cathode interface in sulfide SEs cannot be achieved through a single material or process alone. Instead, it heavily depends on the synergistic optimization of cathode composition ratios, conductive agent types and distribution, composite structure design, and forming processes. By systematically regulating the spatial configuration of ion and electron transport networks at the cathode composition design level, and simultaneously suppressing interfacial side reactions while ensuring high loading capacity and high-rate performance, a solid foundation can be established for achieving high energy density and long-term operation in sulfide-based ASSBs.

4.2.2 Material Coating

Focusing on stabilization design at the cathode/sulfide interface, constructing effective artificial interface layers has emerged as one of the most advanced and engineering-friendly strategies in current research ^[35]. From the perspective of interfacial regulation mechanisms, the core function of introducing a coating layer onto cathode particle surfaces lies in isolating direct contact between sulfide SEs and high-potential CAMs. Simultaneously, by constructing an interface structure that allows ionic conduction while blocking electron flow, it suppresses ongoing oxidation decomposition reactions at the interface. An ideal coating material must satisfy multiple

requirements^[104]. First, good thermodynamic and chemical stability is required at high operating voltages. Second, sufficient lithium-ion conductivity or a low ion migration barrier is necessary to avoid introducing additional interfacial polarization. Third, electronic conductivity should be as low as possible to prevent side reactions induced by electron migration across the interface. Fourth, mechanical flexibility or adaptive deformation capability is essential to mitigate contact degradation caused by volume changes in the CAM during charging and discharging. Additionally, precise control of coating thickness is crucial for performance optimization (**Figure 11a**)^[154]. Excessively thin coatings fail to fully suppress interfacial side reactions, while overly thick layers significantly increase ionic diffusion resistance, leading to diminished rate performance. Within this framework, coating strategies centered on inorganic oxides, phosphates, nitrides, and organic/inorganic hybrid layers have been systematically developed.

Inorganic oxides and inorganic lithium salts were among the earliest materials employed for sulfide cathode interface regulation (**Figure 11b-c**). Coating layers represented by LiNbO_3 ^[111], Li_2CO_3 ^[155], and LiTa_2PO_8 ^[156] exhibit excellent chemical inertness in the high-voltage range. The continuous Li^+ migration pathways within the crystal structure can partially maintain the continuity of ion transport at the interface. These materials are typically uniformly deposited onto cathode particle surfaces via sol-gel, in-situ deposition, or atomic layer deposition methods, constructing continuous, dense artificial cathode-electrolyte interfaces at the nanoscale. Through refined process control, such inorganic coatings have demonstrated significant suppression of interfacial impedance and enhanced cycling stability in high-voltage layered oxide CAMs. Beyond traditional oxides, phosphorus- or nitrogen-containing inorganic amorphous coatings also exhibit unique advantages in sulfide systems (**Figure 11d**)^[157]. Amorphous SE-type coatings, exemplified by LiPON, combine a broad electrochemical stability window with excellent interfacial film-forming capability. When employed as a buffer layer on CAM surfaces, it achieves structural compatibility with sulfide SEs while effectively mitigating interfacial decomposition at high voltages. Compared to crystalline oxides, the amorphous layer lacks long-range ordered

1 structures at the atomic scale, which helps reduce interfacial stress concentration and
2 form a more uniform interfacial composition distribution, thereby enhancing stability
3 during long-term cycling. Additionally, double-layer or multi-layer composite coating
4 structures are gaining attention. By combining inorganic materials with different
5 functions, synergistic optimization can be achieved between thermal stability,
6 electrochemical stability, and ion transport performance ^[112].
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10 At higher levels of interface design, coating strategies have gradually evolved from
11 single-material layers to structured, integrated composite interface engineering (**Figure**
12 **12a**) ^[158]. Unlike traditional inorganic coatings, organic or organic-inorganic hybrid
13 coatings demonstrate unique advantages in accommodating cathode volume expansion
14 (**Figure 12b**) ^[159]. High-nickel layered oxide CAMs undergo significant lattice
15 parameter changes during charge-discharge cycles, readily inducing microcracks on
16 particle surfaces and leading to solid-solid contact degradation. Introducing polymer or
17 conductive polymer coatings with elasticity and adhesion onto the cathode surface
18 creates a flexible buffer zone at the interface, effectively absorbing mechanical stresses
19 induced by volume changes. Simultaneously, through techniques like molecular layer
20 deposition or in-situ polymerization, such coatings achieve highly uniform nanoscale
21 coverage and circumvent incompatibility issues between conventional conductive
22 additives and sulfide SEs (**Figure 12c**) ^[160]. More importantly, the inherent high
23 conductivity of carbon phases can be transformed into semiconductor properties by
24 modulating the electronic structure of the organic layer, fundamentally reducing the
25 possibility of interface side reactions. Furthermore, a continuous electrolyte shell can
26 be formed around cathode particles via mechanical blending, directly constructing a
27 core-shell composite cathode structure. This design significantly shortens the geometric
28 migration path for lithium ions across the interface while eliminating the uneven
29 electrolyte distribution inherent in conventional dry-mixing methods. During cycling,
30 the continuous electrolyte shell undergoes cooperative deformation with the volume
31 changes of cathode particles, maintaining stable solid-solid contact and suppressing the
32 cumulative increase in interfacial impedance. Moreover, the role of the coating layer in
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1 interface stability is not isolated; its effectiveness is often closely tied to the structural
2 stability of the CAM itself. In recent years, strategies combining elemental doping with
3 surface phase regulation have enabled the construction of stable rock salt phases or
4 gradient structures on cathode particle surfaces, significantly mitigating structural
5 evolution behavior in high-nickel cathodes under high voltages (**Figure 12d**) ^[161].
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7 When such structurally stabilized cathodes are synergistically employed with
8 functionalized coating layers, both the chemical reaction driving force and mechanical
9 mismatch at the interface are effectively reduced, thereby achieving multiscale stability
10 enhancement from the bulk material to the interface. Imbalanced electron conduction
11 at the interface is also an important factor inducing cathode-side side reactions. To
12 address this issue, researchers have begun attempting to reconstruct interfacial
13 conduction behavior by regulating the electronic structure of the coating layer ^[162].
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15 Through the introduction of carbon-based or semiconductor coatings with tunable
16 electronic properties, disordered electron transport in the interfacial region is restricted,
17 thereby suppressing the oxidative decomposition of sulfides. 2D graphene-like carbon
18 materials, featuring exceptional mechanical flexibility and controllable conductivity,
19 have been employed to construct interfacial layers that simultaneously provide physical
20 isolation and conductive regulation. Such coatings reduce interfacial contact resistance
21 while precisely controlling the continuity of electronic pathways, thereby decoupling
22 ion-electron transport. This approach significantly enhances interfacial stability without
23 compromising rate performance.
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Regulating the cathode/sulfide interface through material coating has proven to be a critical approach for enhancing the performance and reliability of sulfide-based ASSBs. By rationally selecting coating materials, precisely controlling coating structures, and integrating with CAM modification and battery configuration design, synergistic optimization can be achieved in suppressing interfacial side reactions, reducing interfacial impedance, and mitigating mechanical failure. Future research should focus on investigating the long-term cycling and high-load evolution behavior of the coating layer while developing scalable, low-cost fabrication processes. This will

1 accelerate the practical implementation of sulfide-based ASSBs in high-energy-density
2 and high-safety applications.
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4 *4.2.3 Developing Novel Cathode Materials*

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6 CAMs not only determine the upper limit of the battery's energy density but also
7 profoundly influence the chemical stability, mechanical integrity, and long-term cycling
8 reliability at the cathode/electrolyte interface. Compared to liquid LIBs, sulfide SEs
9 possess high ionic conductivity and excellent mechanical flexibility, yet face relatively
10 limited thermodynamic stability windows at high voltages. When conventional layered
11 oxide cathodes operate under high-voltage conditions, the highly oxidative
12 environment at the cathode surface readily induces irreversible oxidative
13 decomposition of the sulfide SE. This leads to the formation of high-impedance
14 interfacial layers enriched with sulfur, phosphorus, or metal sulfides, subsequently
15 exacerbating interfacial polarization and causing rapid capacity decay ^[130,131]. Driven
16 by this interface failure stemming from intrinsic material mismatch, research focus has
17 gradually shifted from relying solely on interface coating and process optimization
18 toward developing novel CAMs that better suit the sulfide environment.
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33 Recently, novel CAMs centered on non-oxides have emerged as a key direction in
34 sulfide-based ASSB research. These materials primarily include metal halides, metal
35 sulfides, and homogeneous sulfur (sulfur-selenium) compound cathodes with self-
36 supporting mixed transport characteristics. Compared to traditional oxide CAMs, these
37 materials exhibit higher compatibility with sulfide SEs in terms of chemical
38 composition and electronic structure. Their operating voltages typically fall within the
39 relatively stable voltage range of sulfide SEs, significantly reducing the thermodynamic
40 driving force for interfacial oxidation decomposition. Furthermore, non-oxide cathodes
41 generally do not involve the release of reactive oxygen species during reactions, which
42 fundamentally enhances the chemical stability at the cathode/electrolyte interface.
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54 Among numerous novel CAMs, halide cathodes have garnered extensive attention
55 due to the outstanding interface compatibility and tunable structural properties (**Figure**
56 **13**). Material systems represented by lithium iron chlorides, such as $\text{Li}_{1.3}\text{Fe}_{1.2}\text{Cl}_4$ ^[163],
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Li₂FeCl₄ [164], and Li_{2-2x}Fe_{1+x}Cl₄ [165], achieve energy storage through Fe³⁺/Fe²⁺ multi-electron redox reactions. Their moderate reaction potentials effectively circumvent the risk of oxidative decomposition of sulfide SEs at high voltages. Experimental studies demonstrate that these materials achieve high reversible capacity in ASSBs while maintaining a slow interface impedance growth rate during long cycling, indicating excellent interface stability. More importantly, by tuning the Li/Fe ratio in the crystal structure, low-strain or quasi-zero-strain iron-based chloride cathodes can be constructed. This significantly reduces lattice parameter changes during charging and discharging, mitigating contact degradation at the solid-solid interface caused by volume mismatch. These results reveal that designing novel CAMs based on structural strain regulation is an effective approach to enhancing the interface stability of sulfide-based ASSBs.

Beyond iron halides, simpler metal chloride cathodes also demonstrate application potential in sulfide-based ASSBs. Metal chloride cathodes represented by FeCl₃ exhibit limited electronic conductivity but feature simple chemical composition, relatively mature preparation techniques, and mild interfacial side reactions with sulfide SEs in solid-state environments (**Figure 14a**) [166]. Studies indicate that through rational design of composite electrode structures and optimization of particle contact modes, FeCl₃ cathodes can still achieve stable reversible reaction behavior in sulfide-based ASSBs. This finding suggests that achieving high interfacial stability does not necessarily depend on complex multivalent oxide structures, and some metal halides also hold promising development prospects. From a more systematic materials development perspective, comprehensive studies on metal chloride cathodes indicate key advantages in ASSBs: lower reaction potentials, weaker oxidizing properties, and relatively flexible mechanical characteristics (**Figure 14b-c**) [167,168]. These characteristics enable metal chloride cathodes to maintain stable solid-solid interfaces under high packing densities, thereby reducing the accumulation rate of interfacial impedance. Related studies generally agree that metal chloride cathodes offer a universally applicable material design pathway to overcome the interface stability bottleneck of traditional oxide

cathodes in sulfide-based ASSBs.

Additionally, metal sulfide CAMs also play a crucial role in sulfide-based ASSBs. Metal sulfide CAMs, represented by FeS_2 , possess extremely high theoretical specific capacity and excellent resource availability, offering significant advantages in achieving high energy density (**Figure 15a-b**)^[169-171]. Their chemical composition closely resembles sulfide SEs, which confer inherent advantages in interfacial chemical compatibility. However, metal sulfide CAMs typically rely on conversion-type reaction mechanisms, accompanied by pronounced phase transitions and volume changes during charging and discharging. This readily leads to deteriorated interfacial contact, stress concentration, and exacerbated polarization, thereby limiting their long-term stable operation in ASSBs. To address these challenges, researchers have significantly improved the interfacial behavior and cycling stability of metal sulfide CAMs through diverse material design strategies in recent years^[172]. Constructing nanostructures, introducing flexible conductive networks, or performing pre-sulfidation treatment can effectively mitigate volume effects during conversion reactions and enhance reaction kinetics. Research findings indicate that metal sulfide CAMs can achieve relatively stable cycling performance in sulfide-based ASSBs after rational structural and interfacial optimization, laying the foundation for the application of high-capacity conversion cathodes in ASSBs. In the development of novel CAMs, homogeneous sulfur compounds with self-supporting mixed transport properties offer a novel design paradigm for sulfide-based ASSBs (**Figure 15c**)^[173]. By simultaneously establishing continuous electronic and lithium-ion transport pathways within a single crystalline phase, these materials inherently possess hybrid conductivity, greatly reducing reliance on external conductive additives and SEs. Systems exemplified by multi-anionic sulfur-selenium compounds exhibit minimal volume changes and outstanding structural stability during charging and discharging, facilitating long-term maintenance of stable solid-solid interface contact (**Figure 15d**)^[174]. This "cathode homogenization" strategy breaks away from the traditional composite cathode design relying on multiphase interfaces, offering a new direction to fundamentally address internal transport

inhomogeneities and interfacial failures within SEs.

Overall, developing novel CAMs represents one of the fundamental approaches to resolving the cathode/electrolyte interface instability issues in sulfide-based ASSBs. Metal chloride CAMs demonstrate comprehensive advantages in interface stability, structural flexibility, and cost control. Metal sulfide CAMs hold significant potential for high-energy-density applications, while homogenized sulfur (sulfur-selenium) compound cathodes offer a novel approach for integrated optimization of both interface and internal electrode transport. Future research must deepen understanding of the reaction mechanisms, structural evolution, and interfacial behavior of these novel CAMs. This will enable synergistic optimization of material design, interfacial control, and battery architecture, propelling sulfide-based ASSBs toward high performance and practical application.

5 Interface of Sulfide/Anode

5.1 Interface Issues

The electrolyte/anode interface is widely recognized as the core functional region determining battery safety, cycle life, and rate performance. Compared to traditional liquid batteries, the solid-state interface not only performs the essential functions of lithium-ion migration across the interface and charge transfer during charging and discharging but also undergoes multiple coupled processes such as significant mechanical stress evolution, interfacial chemical reactions, and microstructural reconstruction [37,175]. Particularly at the interface between sulfide SE and LMA, the interface exhibits pronounced intrinsic instability due to the highly reactive nature of LMA and the limited electrochemical stability window of sulfide SEs. The failure mechanisms at the sulfide SE/LMA interface can be attributed to three mutually coupled core processes. Firstly, the nucleation and uncontrolled growth of lithium dendrites within the electrolyte or interface ultimately lead to internal short circuits. Secondly, the persistent electrochemical decomposition reaction between LMA and the sulfide SE causes continuous accumulation of high-impedance interfacial phases. Thirdly, stress mismatch and interfacial contact degradation induced by volume

changes during lithium deposition/stripping further amplify current density non-uniformity and dendrite growth tendencies. These mechanisms are not isolated but mutually reinforce and co-evolve during cycling, ultimately obstructing interfacial ion transport pathways. This constitutes the critical bottleneck constraining the practical application of sulfide-based ASSBs.

5.1.1 Nucleation and Growth of Lithium Dendrites

The formation and growth of lithium dendrites represent one of the most destructive processes in the failure of sulfide/LMA interfaces (**Figure 16a**)^[176]. Unlike liquid batteries, where dendrites primarily grow on metal surfaces, lithium dendrites often nucleate directly within SEs and propagate along low-impedance pathways in ASSBs. This constitutes a complex process jointly regulated by electrochemical, mechanical, and microstructural factors^[40]. The evolution of lithium dendrites typically proceeds through two stages: nucleation and growth (**Figure 16b**)^[177]. During the initial nucleation phase, LMA tends to preferentially deposit within subsurface pores, defects, or microcracks near the interface with the sulfide SE. These regions become thermodynamically favorable sites for lithium atom growth due to localized ion flux concentration, electric field distortion, and lower deposition energy barriers. As pores gradually fill with lithium, the volume expansion from continued deposition causes a sharp increase in local pressure and subsequently induces the initiation of new cracks within SE. Upon entering the growth stage, lithium dendrite behavior exhibits the characteristic “wedge-opening” mechanism. Unlike traditional models driven by tip stress concentration, LMA primarily injects continuously from the rear end of the crack, propelling its forward propagation through wedging. When the wedge stress generated by lithium deposition within the crack exceeds the critical fracture toughness of the sulfide SE, the crack undergoes unstable propagation and ultimately penetrates the entire electrolyte layer, leading to catastrophic short-circuit failure.

From an atomic perspective, nucleation sites for lithium dendrites are not limited to the LMA surface (**Figure 16c**)^[178]. Simulation studies indicate that positions within the SEI may also serve as preferred nucleation sites for lithium clusters. Due to the typically

amorphous composite structure of sulfide decomposition products, their bandgap significantly narrows or even exhibits some electronic conductivity. This allows electrons to penetrate the interfacial layer and locally reduce Lithium-ion in situ to LMA, thereby forming electrochemical "dead lithium" clusters. This concealed internal nucleation mechanism greatly increases the unpredictability of dendrite failure. Furthermore, lithium deposition morphology is highly sensitive to current density^[179]. At low current densities, lithium tends to grow in relatively uniform granular or compact forms. Under high current densities, increased local overpotentials drive lithium deposition toward mossy or even dendritic structures. The interfacial phases formed on dendrite surfaces often leave hollow shells during exfoliation. These "dead lithium" structures not only cause active lithium loss but also accumulate continuously at the interface, significantly elevating interfacial impedance.

5.1.2 Synergistic Effects of Microstructural Defects and Electronic Conductivity

The microstructural characteristics of sulfide SEs play a decisive role in lithium dendrite behavior. Extensive studies confirm that impurity particles, grain boundaries, cracks, and local structural inhomogeneities at interfaces significantly increase local current density, serving as preferred initiation sites for lithium dendrite growth. For instance, certain tetrahedral sites and defect regions within the crystal structure of sulfide SEs like Li_3PS_4 exhibit lower lithium deposition energy barriers, facilitating easier reduction deposition in these areas (**Figure 17a**)^[180]. More critically, sulfide SEs exhibit non-negligible intrinsic electronic conductivity, particularly pronounced in defect-rich regions or after decomposition product formation. This electronic conductivity enables lithium ions and electrons to undergo recombination reactions within the electrolyte, resulting in the spontaneous nucleation of lithium nuclei far from the LMA and subsequent sustained growth along low-impedance pathways (**Figure 17b**)^[181]. In polycrystalline materials, grain boundaries and intergranular cracks provide ideal expansion pathways for Lithium dendrites. The presence of surface electronic conductivity further distorts the electric field distribution, driving lithium migration along specific paths and significantly reducing the critical current density required for

dendrite penetration (**Figure 17c-d**)^[182].

5.1.3 Mechanical Mismatch and Degradation

Beyond electrochemical factors, mechanical degradation represents another critical root cause of instability at the sulfide SE/LMA interface. Lithium undergoes significant volume changes during deposition and stripping, while sulfide SEs typically exhibit brittle fracture characteristics. This vast disparity in mechanical properties inevitably leads to stress concentration at the interface. During cycling, lithium deposition imposes sustained compressive stress at the interface, whereas the delithiation phase generates interfacial voids and contact loss due to lithium volume contraction^[183]. This cyclic stress loading not only accelerates the propagation of pre-existing cracks within the electrolyte but also induces mixed-mode fracture at grain boundaries, thereby creating new pathways for lithium dendrite growth (**Figure 18a-b**)^[184,185]. Simultaneously, electrochemical decomposition reactions at the interface generate interfacial phases with mixed ion-electron conductivity. This enhanced electron conductivity further accelerates electrolyte degradation, forming a typical “mechanical-chemical-electrical” coupled failure pathway. The magnitude of the critical current density is fundamentally determined by the interfacial density, the ionic conductivity of the electrolyte, and its mechanical integrity. Under high current density conditions, lithium deposition morphology becomes finer and more prone to forming whisker or dendritic structures. During the delamination stage, volume discontinuities caused by the inability to promptly backfill vacancies continuously amplify local current density and accelerate the failure process (**Figure 18c**)^[186].

5.1.4 Interface Evolution

As electrochemical cycles progress, the sulfide SE/LMA interface does not remain static but undergoes continuous structural and chemical evolution^[187]. When the lithium stripping rate exceeds the vacancy diffusion threshold, the accumulation of vacancies at the interface inevitably leads to the formation of micrometer-scale voids. This results in a reduction of the interface contact area and a sharp increase in polarization. Notably, lithium vacancy diffusion is synergistically driven by both

concentration gradients and stress fields, leading to a more uneven interface current distribution that induces new deposition hotspots (**Figure 19a**) ^[188]. By applying moderate stacking pressure, LMA can be induced to undergo elastic or creep deformation, maintaining tight interfacial contact and delaying void formation (**Fig. 19b-d**) ^[189-191]. However, excessively high external pressure may drive lithium to permeate along electrolyte grain boundaries, thereby increasing the risk of dendrite growth and structural damage. Consequently, there exists a well-defined window range for interfacial pressure regulation. Furthermore, the strong reductive nature of LMA makes it thermodynamically difficult for most sulfide SEs to coexist with it. Decomposition reactions inevitably produce interface products with low ionic conductivity, leading to the continuous accumulation of interfacial impedance ^[107]. The presence of an electron-conductive phase within the decomposition products further triggers chain reactions, accelerating interfacial failure and even thermal runaway.

5.2 Strategies for Modifying the Anode Interface

5.2.1 Optimization of Anode Materials

Carbon-based anodes, particularly graphite materials, remain widely regarded as the most feasible anode option for the early commercialization stage of ASSBs due to the mature industrial infrastructure, stable layered structure, and excellent cycling reliability (**Figure 20a**) ^[192]. From an engineering perspective, graphite offers significant advantages in material cost, resource availability, and processing techniques. Its minimal volume change also ensures good structural stability within solid-state systems. However, when graphite anodes come into direct contact with sulfide SEs, the interface stability is significantly inferior to that of traditional liquid batteries. This issue has become a key challenge limiting the application of graphite in sulfide-based ASSBs ^[193]. The fundamental cause of this interface instability lies in the relatively narrow electrochemical stability window of sulfide SEs ^[22]. Under low-potential conditions, sulfide anions in the electrolyte readily undergo reduction decomposition, yielding decomposition products that are electron insulators with limited ionic conductivity ^[74]. Such interfacial decomposition layers typically exhibit complex chemical compositions,

disordered structures, and uncontrollable growth characteristics. This makes it difficult to achieve the self-passivation effect seen in stable SEI layers within liquid batteries, leading to continuously increasing interfacial impedance and accelerated capacity decay.

Further studies reveal that the amorphous carbon layer prevalent on graphite surfaces exhibits a significant amplification effect on interfacial reactions within ASSBs^[194]. Due to its high specific surface area and abundant defect states, amorphous carbon readily undergoes electron transfer reactions with sulfide SEs, thereby accelerating the reduction decomposition process of the electrolyte. In contrast, highly crystalline graphite obtained by removing amorphous carbon through chemical or thermal treatment shows markedly improved interface stability when forming solid-state interfaces with sulfide SEs (**Figure 20b**)^[195]. The surface of highly crystalline graphite effectively suppresses the ongoing decomposition of the electrolyte, inducing the formation of a thinner, denser, and more ionically conductive interface layer. This significantly reduces the interfacial charge transfer resistance and enhances cycling stability. It is noteworthy that in ASSBs, a reduction in electrolyte content often amplifies interfacial issues^[196]. However, highly crystalline graphite maintains stable electrochemical performance even under conditions of significantly reduced sulfide SE proportion. This indicates that interfacial chemical stability takes precedence over ionic conductivity, playing a decisive role in the overall performance of ASSBs. Furthermore, due to the lack of "wetting" and "self-healing" capabilities for interfacial defects present in liquid batteries, carbon-based anodes are more prone to exposing their interfacial kinetic weaknesses under high-rate or rapid charge-discharge conditions. To address these issues, introducing functional coatings or engineered interfacial layers onto the graphite surface can modulate the interfacial electronic structure and suppress side reactions (**Figure 20c**)^[197]. Such interfacial engineering strategies block electron hopping while maintaining efficient lithium-ion transport, thereby establishing a more stable solid-solid contact interface between sulfide SEs and carbon-based anodes. Overall, the core challenge for carbon-based anodes in sulfide-based ASSBs lies not in dendrite growth, but in suppressing electrolyte reduction decomposition at low

potentials while achieving long-term stable interfacial chemical and electrochemical behavior.

Silicon-based anodes are regarded as key candidates for next-generation high-energy-density anode materials due to the exceptionally high theoretical specific capacity and relatively low tendency for lithium dendrite formation^[198]. However, the interface stability issues encountered by silicon anodes in sulfide-based ASSBs are primarily driven not by electrochemical decomposition, but by the dramatic volume expansion behavior during charging and discharging. Silicon undergoes volumetric expansion of up to approximately 300% during lithiation, and this substantial strain is difficult to accommodate through structural rearrangement or interfacial slip under solid-state conditions. Consequently, severe mechanical mismatch is induced at the anode/electrolyte interface^[199]. In situ structural characterization and multiscale simulation studies systematically revealed the crack evolution behavior of silicon-based anodes in ASSBs (**Figure 21a**)^[200]. During charge-discharge cycles, multiple crack structures progressively form within the silicon electrode, including channel-type cracks propagating along the electrode thickness, delamination cracks at interfaces due to stress concentration, and microcrack networks that periodically open and close during cycling. These cracks not only significantly reduce the effective contact area between the anode and sulfide SE but also cause highly non-uniform local current density and stress field distributions, thereby accelerating the growth of interfacial impedance and inducing rapid battery performance degradation. The development of crack networks is highly correlated with the evolution of battery internal resistance, while electrode thickness uniformity and initial stress state are key factors determining crack evolution pathways and severity. Relatively uniform electrode structures help disperse stress concentrations and delay interface failure, whereas thickness variations or locally densified regions are more prone to serve as crack initiation sites.

From the perspective of multi-level structural optimization, the interfacial stability of silicon-based anodes is simultaneously influenced by the synergistic effects of particle scale, interfacial scale, and electrode scale (**Figure 21b**)^[201]. At the particle

level, the intrinsically low electronic conductivity and long lithium-ion diffusion paths of silicon materials can induce severe polarization under high-rate conditions. Therefore, nanostructured designs or heteroelement doping are required to shorten diffusion distances and enhance conductivity. At the interface level, unstable reaction products often exhibit high ionic migration impedance. Constructing engineered interface layers or introducing buffer coatings can improve interfacial conductivity and chemical stability. At the electrode level, changes in porosity and crack propagation caused by volume expansion during cycling significantly lengthen ion transport paths. This necessitates maintaining overall electrode structural integrity by optimizing particle size distribution, introducing functional binders, and enhancing the mechanical coupling between the current collector and active layer. Although silicon exhibits far superior theoretical capacity compared to graphite, its interfacial mechanical stability issues are particularly pronounced in sulfide-based ASSBs. Current research generally agrees that the direct application of pure silicon anodes in ASSBs remains challenging. However, constructing composite anode systems with low-expansion materials like graphite holds promise for balancing capacity enhancement and interface stability [202].

In the pursuit of higher energy density, lithium alloy anodes have gradually emerged as a crucial transitional solution bridging traditional graphite anode and LMA (**Figure 22a**) [203,204]. Compared to pure LMA, lithium alloy anodes mitigate the risk of dendrite growth while enabling lithium storage through reversible alloying/dealloying reactions [205]. However, lithium alloy anodes within sulfide-based ASSBs also face complex interface stability issues stemming from volume changes. Research indicates that alloy anodes typically undergo significant crystalline-amorphous transitions and irreversible volume expansion during the first cycle, accompanied by pronounced stress hysteresis effects and structural rearrangement behavior [206]. As cycling progresses, interparticle rearrangement and adaptive adjustments to the pore structure partially alleviate stress concentration, gradually stabilizing overall stress levels. However, this stress self-regulation process often requires sacrificing structural integrity and may still induce contact degradation at interfaces (**Figure 22b**) [207].

1 The size of alloy particles decisively influences interfacial stability ^[208]. Smaller
2 particles can more effectively utilize internal pore space to buffer volume changes,
3 thereby reducing interfacial stress concentration. Conversely, larger particles tend to
4 induce overall deformation of the entire electrode during lithiation, which rapidly
5 deteriorates interfacial contact. Furthermore, external stacking pressure plays a crucial
6 role in regulating the interface stability of alloy anodes. The pressure applied in ASSBs
7 must match the intrinsic yield strength of the alloy material to achieve a balance
8 between interface densification and structural stability. Insufficient pressure will lead
9 to void formation and contact loss, whereas excessive pressure may induce irreversible
10 plastic deformation and accelerate material degradation (**Figure 22c**) ^[209,210]. To
11 address these challenges, constructing pre-lithium alloy systems or introducing buffer
12 phases with low Young's modulus can partially absorb volumetric stress and stabilize
13 the interface structure (**Figure 22d**) ^[211]. Such strategies not only reduce interfacial
14 impedance but also enhance overall coulombic efficiency by compensating for active
15 lithium loss during cycling, demonstrating promising application potential in sulfide-
16 based ASSBs.

17 By replacing traditional anode materials with a metallic current collector, lithium-
18 free ASSBs enable in-situ deposition of LMA during charging, theoretically achieving
19 higher mass and volumetric energy densities. This design concept not only eliminates
20 processing and safety hazards associated with preformed LMA but also significantly
21 simplifies battery manufacturing processes (**Figure 23a**) ^[212,213]. However, in sulfide-
22 based ASSBs, the interface stability of lithium-free anodes critically depends on the
23 reversibility and uniformity of lithium deposition/dissolution. During charging, lithium
24 ions are extracted from the cathode and electrochemically deposited onto the current
25 collector surface, forming an active lithium layer. During discharging, this layer must
26 completely dissolve and redeposit within the cathode. If lithium deposition is non-
27 uniform, it readily forms high-surface-area structures that gradually evolve into
28 electrochemically inert dead-lithium during cycling, continuously depleting the limited
29 lithium supply. Since lithium in lithium-free anode batteries originates entirely from the

cathode, this issue is particularly critical in ASSBs. Research indicates that introducing a functional interlayer between the current collector and SE can effectively regulate lithium nucleation and growth behavior (**Figure 23b**)^[214,215]. Interlayer materials with reversible alloying capability induce uniform lithium nucleation at the interface and form low-curvature deposition morphologies, thereby alleviating stress concentration and suppressing dendrite growth. Furthermore, the interlayer partially functions as an electron buffer layer, reducing the probability of direct electron injection into the sulfide SE and thereby enhancing interfacial chemical stability. However, lithium-free anode ASSBs exhibit extreme sensitivity to coulombic efficiency. Their cycle life is not only governed by interfacial reactions at the anode but also closely related to irreversible reactions at the cathode (**Figure 23c**)^[216]. When the capacity ratio between cathode and anode approaches 1, even minor irreversible lithium loss is rapidly amplified and becomes a decisive factor limiting battery lifespan. Therefore, when evaluating the interface stability of lithium-free anodes in sulfide-based ASSBs, it is essential to establish a comprehensive analytical framework that encompasses cathode-anode synergistic effects to fully dissect the contribution of different irreversible reactions to overall performance.

5.2.2 Modification of Sulfide SEs

From a thermodynamic perspective, most sulfide SEs possess relatively narrow intrinsic electrochemical stability windows and are prone to reductive decomposition reactions when directly contacting LMA under low-potential conditions. This interfacial instability typically manifests as continuous electrolyte decomposition, increasing interfacial impedance, and uncontrolled lithium deposition morphology, which subsequently induces lithium dendrite nucleation and propagation. These factors severely limit battery cycle life and safety. Therefore, simply relying on the intrinsic properties of materials is insufficient to achieve a long-term stable LMA interface. This has prompted researchers to focus on sulfide SE material design, employing strategies such as composition modification, microstructure optimization, and composite formation to in situ construct a stable SEI layer at the LMA interface. Doping

heteroatoms to modulate the composition and electronic structure of sulfide SEs represents an effective approach to enhance negative electrode interface stability (**Figure 24a-b**) ^[217,218]. By altering the local coordination environment, regulating carrier migration behavior, and inducing the restructuring of interface reaction products, doped elements significantly influence the kinetics of interface reactions between the electrolyte and LMA. Research indicates that halogen element doping in sulfide SEs exhibits multiple beneficial effects (**Figure 24c**) ^[68,219]. On one hand, halogen introduction effectively enhances the ionic conductivity of sulfide SEs, reduces concentration polarization at the interface, and homogenizes lithium-ion flux distribution from a kinetic perspective. On the other hand, halogen-enriched regions readily form in-situ lithium halide interfacial layers upon contact with LMA. Such layers typically exhibit high electronic insulation and excellent ionic conductivity, blocking electron permeation into the bulk electrolyte and thereby suppressing persistent reduction decomposition reactions. From the perspective of interfacial evolution mechanisms, lithium dendrite formation and propagation fundamentally stem from the dynamic instability of the SEI layer on the anode surface ^[220]. Therefore, inducing the formation of an interfacial phase with high mechanical strength and electrochemical stability through sulfide SE material design is a core objective for achieving long-term stable LMA interfaces. By finely modifying sulfide SEs, the in-situ construction of stable fluoride interface layers on electrolyte surfaces emerges as an effective strategy. The introduction of fluorine not only enhances the chemical stability of the interfacial phase but also significantly reduces the reduction decomposition tendency of sulfide SEs. The in-situ formed fluoride layer typically exhibits high interfacial energy and excellent electronic insulation, guiding uniform LMA deposition while suppressing dendrite nucleation. This significantly enhances the cycling stability of ASSBs (**Figure 24d**) ^[97,221].

Beyond static interface regulation, constructing interface structures with dynamic adjustment capabilities is considered a higher-order strategy for suppressing lithium dendrite growth. By introducing components with high lithium-ion mobility and good

lithium affinity into sulfide SEs, a dynamic interface layer with self-healing properties can be induced at the electrolyte/LMA interface. Research indicates that incorporating iodide components such as LiI into sulfide SEs can form a nanoscale LiI interface layer during lithium deposition/stripping (**Figure 24e**)^[222]. This interface layer undergoes reversible diffusion and restructuring during electrochemical processes, continuously homogenizing the lithium-ion flux distribution at the interface and significantly reducing local current density peaks. Concurrently, the inherently high ionic conductivity and excellent lithium affinity of LiI contribute to building a stable and continuous SEI structure, thereby suppressing dendrite nucleation and retarding its growth. Furthermore, the favorable plasticity exhibited by LiI enables it to effectively fill the microscopic voids between sulfide SE particles, improving solid-solid contact quality and reducing interfacial contact impedance (**Figure 24f**)^[41]. This "dynamic-mechanical-electrochemical" multi-synergistic mechanism enables iodine-doped sulfide SEs to exhibit outstanding long-term stability at the LMA interface. Introducing sulfide SEs into multiphase systems by constructing ceramic/polymer composite SEs represents another crucial approach to enhancing anode interface stability^[223]. In such SEs, the sulfide primarily facilitates efficient lithium-ion transport, while the polymer utilizes its flexible properties to buffer interfacial stresses and improve contact. Compared to conventional mechanical mixing methods, employing chemical bonding strategies to uniformly anchor the sulfide SE within the polymer matrix effectively eliminates interfacial phase separation and space charge effects, establishing continuous lithium-ion transport pathways. The strong interfacial bonding formed between the sulfide and polymer stabilizes the electrode/electrolyte interface structure and suppresses localized lithium-ion flux concentration. The synergistic combination of the high ion conductivity of the sulfide SE and the mechanical flexibility of the polymer phase in the composite system promotes more uniform lithium deposition at the interface, significantly reducing the risk of dendrite growth.

5.2.3 Constructing the SEI at the Interface of Sulfide SE/Anode

In sulfide-based ASSBs, the interface stability between the LMA and the SE is

1 widely recognized as one of the core factors determining battery safety, rate
2 performance, and cycle life. Compared to the dynamic SEI layer spontaneously formed
3 through solvent decomposition in liquid electrolytes, sulfide SEs often undergo more
4 intense and irreversible solid-solid chemical reactions when directly contacting LMA.
5 The resulting interfacial byproducts typically exhibit both electronic and ionic
6 conductivity, triggering continuous decomposition, a rapid increase in interfacial
7 impedance, and lithium dendrite penetration failure. Therefore, constructing an
8 artificial SEI that combines high lithium-ion conductivity, intrinsic electronic insulation,
9 and excellent mechanical stability has become a core research direction for sulfide-
10 based ASSBs [72,104]. From the fundamental principles of interface regulation, an ideal
11 SEI should possess a nanoscale thickness to avoid significantly increasing interfacial
12 polarization. Simultaneously, this interface layer must provide continuous, low-barrier
13 lithium-ion migration pathways to support reversible lithium deposition and dissolution
14 under high-rate conditions. Crucially, the interface layer should exhibit intrinsic
15 electronic insulating properties, effectively suppressing electron leakage to reduce the
16 effective chemical potential of LMA. This confines lithium within the electrochemical
17 stability window of the SE, thereby preventing further reduction and decomposition of
18 the electrolyte [114]. Furthermore, the interfacial layer must possess sufficient structural
19 integrity and mechanical adaptability to accommodate the inevitable volumetric
20 changes of LMA during cycling. Based on these design objectives, interfacial
21 engineering strategies centered on inorganic interlayers, polymer composite layers, and
22 multiphase heterogeneous structures have garnered significant attention in recent years
23 [224].

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25 Among various interface design approaches, introducing inorganic or inorganic
26 composite interlayers to regulate electronic and ionic transport behavior at interfaces is
27 one effective strategy for achieving interface stabilization. Relevant studies indicate
28 that incorporating inorganic phases with low electronic conductivity and high chemical
29 stability into the interface of sulfide SEs and LMA can effectively suppress electron
30 penetration without significantly compromising ionic transport properties, thereby

mitigating the occurrence of interfacial side reactions. By compositing micron-scale sulfide SEs with nanoscale oxide or phosphate fast ion conductors, a synergistic multiphase structure can be formed at the interface (**Figure 25a**)^[225]. In this structure, the oxide phase primarily serves as an electron barrier and provides chemical stabilization, while the sulfide phase maintains overall high lithium-ion conductivity. Such composite interfacial layers not only reduce interfacial electronic conductivity but also homogenize local current density through multiphase interface construction, thereby suppressing initial nucleation of lithium dendrites. This inorganic composite interfacial layer strategy lays a crucial foundation for the stable operation of sulfide SEs under high current density and long cycling conditions. Beyond inorganic interface layer design, polymer interface layers exhibit unique advantages in mitigating interfacial stress concentration and enhancing contact stability due to excellent flexibility and processability^[226]. For the chemical incompatibility between SEs and LMA, researchers achieved synergistic optimization of interfacial chemical stability and ion transport performance by constructing ultrathin polymer protective layers on electrolyte surfaces. Through molecular structural design and in-situ cross-linking reactions, polymer interfacial layers can form dense and uniform network structures. These not only effectively eliminate side reactions triggered by active functional groups but also provide mechanical buffering during LMA deposition/stripping. More importantly, a rationally designed polymer interface layer retains strong lithium-ion solvation and migration capabilities, thereby avoiding a significant increase in interfacial impedance. This demonstrates the irreplaceable role of the polymer interface layer in enhancing both chemical inertness and mechanical adaptability at the interface.

Building upon this foundation, strategies for constructing functionalized SEI through in-situ interfacial reactions have garnered increasing attention. By introducing a sacrificial layer or mixed-conductor interfacial layer between LMA and sulfide SE, specific structural and functional interfacial products can be generated through controlled reactions during the initial contact phase (**Figure 25b**)^[227]. Upon contact with LMA, halides or metal salts spontaneously form lithium alloy frameworks and

lithium-rich ion-conductive phases, thereby establishing continuous lithium-ion transport pathways at the interface. Simultaneously, the stable salts formed through reaction serve as interfacial buffer phases, reducing the chemical potential difference between LMA and sulfide SE while mitigating the progression of interfacial side reactions. This interfacial engineering strategy achieves synergistic regulation of thermodynamic and kinetic behavior at the interface through the principle of "trading reaction for stability". Furthermore, the concept of hybrid conductor interfacial layers offers novel insights for fine-tuning interfacial electron and ion transport (**Figure 25c**)^[228]. By incorporating inorganic phases with high lithium-ion conductivity and carbon-based or nitride phases with tunable electron transport capabilities into the interfacial layer, a controlled electrochemical potential gradient can be established at the interface. On one hand, the high-ionic-conductivity phase helps reduce local current density and guides uniform lithium-ion migration. On the other hand, the restricted electron transport capability suppresses disordered electron diffusion at the interface, thereby lowering the probability of lithium dendrite nucleation. This interfacial layer enables rapid lithium-ion transport while effectively preventing continuous interface decomposition caused by electron leakage, ensuring stable cycling under high-rate conditions. Beyond chemical stability and electrochemical performance, the significant volume changes accompanying LMA during charging and discharging impose heightened demands on the integrity of the interface structure. Addressing this challenge, biomimetic interface design and elastic interfacial layer strategies have emerged as research hotspots. By constructing inorganic or inorganic-organic composite SEI layers with strong chemical bonding capabilities on the LMA surface, robust adhesion and uniform coverage of the interface layer can be achieved, effectively suppressing electrolyte permeation and dendrite penetration (**Figure 25d**)^[229]. Simultaneously, interfacial layers enriched with flexible polymer components dynamically respond to the volume expansion and contraction of LMA, mitigating stress concentration-induced interfacial delamination. Such interfacial layers exhibit adaptive and self-healing capabilities during cycling, contributing to the maintenance

of long-term stable ion transport pathways (**Figure 25e**)^[230].

Among various interfacial components, LiF-rich SEI is widely recognized as a key factor for achieving high interfacial stability. Although bulk LiF exhibits low ionic conductivity, its low lithium-ion diffusion energy barrier and high interfacial energy at the nanoscale enable it to effectively guide the uniform distribution of lithium-ion flux. Artificially constructed or in-situ reaction-generated LiF-rich interfacial layers can form self-limiting electronic barrier structures at the interface, thereby suppressing the occurrence of persistent side reactions (**Figure 26a**)^[231,232]. Further integration with composite designs featuring high ionic conductivity phases can significantly enhance interfacial ionic transport efficiency while ensuring stability, highlighting the crucial role of interfacial energy regulation in SEI design (**Figure 26b-c**)^[233,234]. Based on research into LiF-rich SEI layers, the construction of gradient interface layers provides a more universal design paradigm for regulating anode interface stability (**Figure 26d-f**)^[235-237]. Unlike homogeneous interfacial layers, gradient interfacial layers introduce continuous variations in chemical composition, electronic/ionic conductivity, and mechanical properties perpendicular to the interface. This achieves synergistic mitigation of multiple mismatch issues between LMA and SE. Typically, gradient interfacial layers comprise a lithium alloy layer adjacent to the LMA and a lithium salt layer adjacent to the SE. The lithium alloy layer effectively suppresses non-uniform lithium deposition and initial dendrite nucleation by lowering the effective chemical potential at the LMA surface and enhancing interfacial mechanical strength. The lithium salt layer, possessing excellent electronic insulation and high lithium-ion mobility, serves as a stable ion-selective transport medium, which further blocks electron permeation and inhibits the reductive decomposition of the SE. Beyond material design, regulating the interfacial contact state is equally crucial for enhancing the stability of the SE/LMA interface. Appropriate stacking pressure helps maintain tight solid-solid interface contact, reduces contact impedance, and suppresses void formation^[238]. When applied pressure falls below the critical threshold, LMA stripping rates may exceed the mechanical compensation capacity of the interface, leading to

1 contact failure and a sharp increase in polarization. Within a reasonable pressure range,
2 the creep behavior of LMA can promote interface re-adhesion, enabling reversible
3 regulation of interface impedance. Introducing flexible buffer layers or highly
4 deformable SEs enables stable interface contact under lower pressure conditions,
5 thereby avoiding material fatigue and interfacial side reactions caused by excessive
6 pressure [239].
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12 Overall, researchers have proposed systematic solutions for constructing the SEI at
13 the interface of sulfide SE/anodes from multiple dimensions, including inorganic
14 interfacial layers, polymer composite layers, mixed conductor structures, LiF-rich
15 interfaces, and pressure-synergistic regulation. These strategies achieve multiscale
16 synergistic optimization between interfacial chemical stability, ion transport kinetics,
17 and mechanical adaptability, laying a crucial foundation for the long-term stable
18 operation of sulfide-based ASSBs. In the future, by deepening the understanding of
19 interfacial reaction mechanisms and structural evolution patterns, along with
20 integrating scalable interfacial engineering processes, it is anticipated that sulfide-based
21 ASSBs will advance toward high energy density and engineering applications.
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33 **6 Summary and Future Prospects**

34 ASSBs are widely regarded as the core technological pathway to overcome the
35 energy density and safety limitations of existing LIBs, with SEs serving as the critical
36 material system determining their performance bottlenecks and engineering feasibility.
37 This review focuses on sulfide-based ASSBs, systematically examining stability issues
38 in sulfide SEs from intrinsic material properties to electrode interface behavior. It
39 summarizes recent significant research advances across multiple dimensions, including
40 material design, interface engineering, and structural regulation. A comprehensive
41 analysis of the inherent instability of sulfide SEs and the failure mechanisms at their
42 interfaces with cathodes or anodes reveals that while sulfide SEs confer high ionic
43 conductivity and excellent formability to ASSBs, they also introduce a series of cross-
44 scale, strongly coupled stability challenges. These issues collectively determine the
45 difficulties and pathways for sulfide-based ASSBs to transition from the laboratory to
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practical applications.

From the material perspective, sulfide SEs can achieve high lithium-ion conductivity comparable to liquid electrolytes at RT, making them one of the SE types closest to meeting industrialization requirements. However, their structural characteristics, dominated by covalent-ionic hybrid bonds, render them highly chemically unstable in air, moisture, and organic solvent environments. They are prone to hydrolysis, oxidation, or solvent-induced decomposition reactions, leading to issues such as H₂S release, decay in ionic conductivity, and interfacial side reactions. Additionally, under operating conditions such as high temperatures and voltages, the thermal stability and electrochemical stability window of sulfide SEs are limited, making it difficult to fully meet the operational demands of high-voltage CAMs and LMA. More critically, sulfide SEs generally exhibit low fracture toughness and limited plastic deformation capacity. During battery cycling, microcracks and structural damage readily form in stress-concentration zones, creating potential pathways for interfacial failure and lithium dendrite penetration. Consequently, the stability issues of sulfide SEs stem not from a single material defect but from the combined effects of chemical, electrochemical, thermal, and mechanical factors.

At the sulfide/cathode interface, the chemical and electrochemical incompatibility between high-voltage CAMs and sulfide SEs represents one of the core obstacles limiting energy density enhancement. Under high-voltage conditions, sulfide SEs readily undergo oxidative decomposition, forming high-impedance interfacial layers that exacerbate interfacial polarization and accelerate capacity decay. Moreover, structural evolution of CAMs, transition metal dissolution, and oxygen release behavior under high voltage and high state-of-charge conditions further accelerate interfacial reactions and induce chemically-mechanically coupled failure. To address these challenges, researchers have proposed multiple cathode interface regulation strategies, including optimization of cathode composition and microstructure, surface coating with stabilizing layers, and development of novel CAMs with inherently superior interface compatibility. While these approaches have mitigated interfacial side reactions and

improved cycling stability, their long-term effectiveness, process complexity, and compatibility with high-loading CAMs require further validation.

In contrast, the stability issues at the sulfide/anode interface are more complex and challenging. On one hand, the high chemical reactivity and dynamic volume changes of LMAs make them highly susceptible to reductive decomposition reactions with sulfide SEs, forming mixed-conductor interfacial layers. This leads to continuous interface evolution, increasing impedance, and even short-circuit failures. On the other hand, microstructural defects, localized conductivity inhomogeneities, and limited mechanical strength in sulfide SEs synergize with lithium deposition behavior, significantly promoting lithium dendrite nucleation and growth. Research addressing these challenges has focused on anode material optimization, sulfide SE composition and structural modification, and the formation of stable SEI layers. Notably, the development of inorganic/organic composite interfacial layers, LiF- or Li₃N-rich interfaces, and gradient-structured interfaces has demonstrated significant advantages in blocking electron conduction, homogenizing lithium-ion flux, and mitigating mechanical mismatch. These approaches provide crucial insights for achieving high current density and long-term cycling stability.

The core scientific questions surrounding sulfide-based ASSBs have shifted from "whether it is feasible" to "how to achieve long-term stable and efficient operation". However, to truly advance ASSBs toward large-scale application, substantial breakthroughs are still required in the following areas.

1) At the material level, the shift from "passive stabilization" to "intrinsic stability" is necessary. Current research predominantly relies on interfacial coating or external protective layers to mitigate the instability of sulfide SEs. However, these strategies often increase the complexity of the material system and may introduce new interfacial issues. Future efforts should prioritize enhancing the intrinsic stability of sulfide SEs in air, electrochemical, and mechanical environments through composition design, structural regulation, and bonding optimization. This may involve multi-anion synergistic regulation, locally designed hybrid rigid-flexible structures, and low-energy

defect engineering to fundamentally reduce reaction driving forces.

2) From the interface perspective, it is essential to transition from understanding static interfaces to dynamic regulation. Extensive experiments demonstrate that the interface between sulfide SEs and electrodes is not static but continuously reconfigures under the combined effects of electrochemical cycling, stress evolution, and chemical reactions. Future research must integrate in-situ characterization techniques with multiscale simulation methods to systematically uncover the temporal and spatial evolution patterns of interfaces. This will clarify the intrinsic connections between different failure modes, providing predictable and quantifiable theoretical foundations for interface design.

3) At the structural and process level, material properties must be considered in combination with battery manufacturing conditions. Sulfide SEs exhibit high sensitivity to pressure, packing density, and micro-contact states, while the outstanding performance achieved under laboratory conditions often relies on elevated stacking pressures or idealized interfaces. Future efforts should explore structural designs and process routes under low-pressure or even no external pressure conditions while ensuring interface stability, thereby meeting the demands of practical battery manufacturing and long-term operation.

4) From the perspective of system integration and application, the evaluation system for sulfide-based ASSBs urgently requires standardization. Significant variations in electrolyte thickness, electrode loading, test pressure, and cycling conditions across studies have resulted in a lack of unified metrics for performance comparison. Establishing testing protocols and failure criteria tailored to real-world application scenarios will facilitate objective assessment of the true potential of diverse materials and interface strategies.

In summary, significant progress has been made in material performance and interfacial control for sulfide-based ASSBs, but stability issues remain the core bottleneck hindering industrialization. Future research must establish an effective closed-loop system integrating intrinsic material stability enhancement, dynamic

interfacial regulation mechanism analysis, and engineering feasibility. Through interdisciplinary and multi-scale collaborative innovation, sulfide-based ASSBs can advance from proof-of-concept to practical application with high reliability and consistency.

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Conflicts of Interest

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